

Sell Now or Keep a Share? Risk-Adjusted Optimization of Academy Transfer Structures in Football

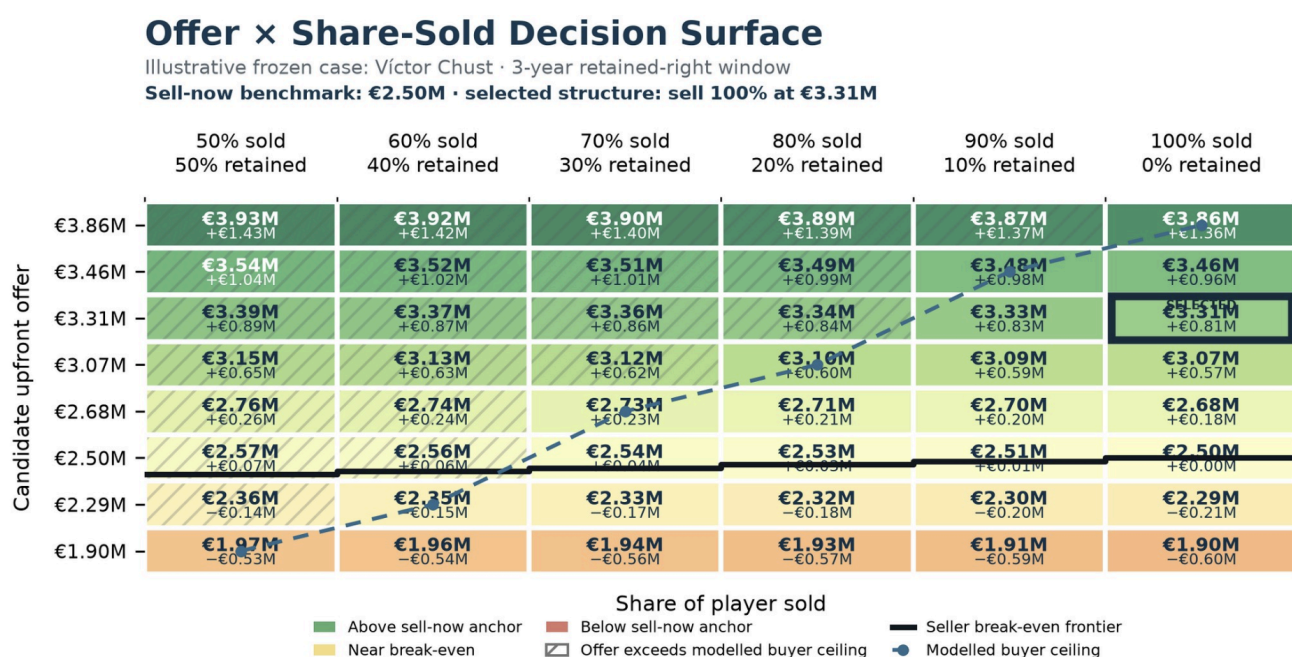
Soccer

Paper ID: [To be assigned]

Abstract.

For an academy graduate, a transfer can involve more than simply selling or keeping the player. A club may take cash today or preserve exposure to future value. We formalize that trade-off as an offer-by-share decision. Each candidate structure must satisfy the seller's break-even requirement and a modelled buyer ceiling. We apply the framework to 36 modellable Real Madrid academy exits.

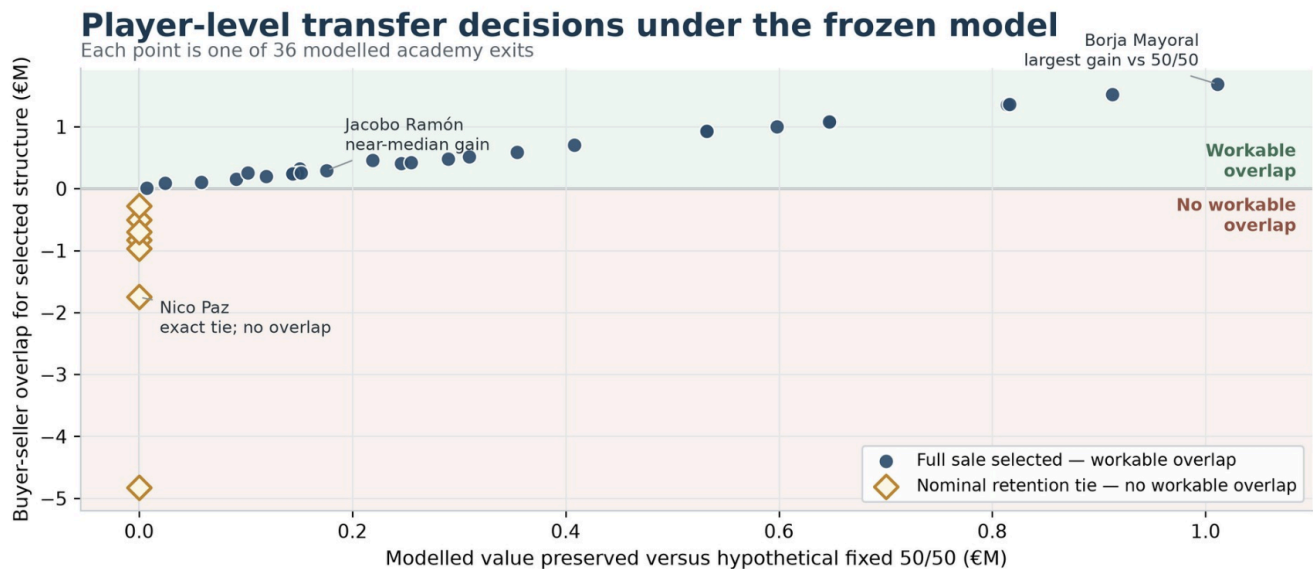
Using decision-date information from roughly 9,000 historical valuation trajectories, we generate one-, two-, and three-year value forecasts. The one- and two-year horizons use persistence, while the three-year horizon uses a 400-tree Extra Trees regressor trained on a log-ratio target. In chronological out-of-sample testing, three-year mean absolute error falls from €21.91M for the preceding ridge model to €13.54M. These forecasts then enter a grid search covering 0-80% retained share and validity windows from one to five years. Each candidate structure accounts for discounted future-value scenarios, transfer probability, duration burden, buyer costs, a target-margin buffer, and downside risk.



Cells are seller-side risk-adjusted modelled values, not realised proceeds. Buyer ceilings are modelled, not observed acceptance.

Figure 1. Offer × share-sold decision surface for an illustrative player case. Each cell combines a candidate upfront offer with the risk-adjusted retained component and reports the difference from the €2.50M sell-now benchmark. The solid staircase marks seller break-even; the dashed line is the modelled buyer ceiling, and hatching marks offers above it. The outlined cell is model-selected. Values are modelled, not realised proceeds or observed buyer acceptance.

Under the default player-specific policy, 29 cases resolve to full sale and seven to nominal retention structures. Those seven retention choices are numerical ties with full sale: they add €0 in modelled value and do not produce workable buyer-seller overlap. Across the portfolio, player-specific selection and full sale therefore both total €172.37M. By comparison, a hypothetical fixed 50/50 policy totals €161.65M. The player-specific framework preserves €10.72M, or 6.63%, equivalent to about €297,860 per modelled exit, relative to that mechanical rule. The gain comes from avoiding indiscriminate 50/50 retention rather than from retained shares outperforming full sale.



Decision message: sell all 29 workable cases; treat the seven retention choices as numerical boundary ties, not value-creating deals.

Figure 2. Cohort-level decisions. All 29 workable cases select full sale; seven nominal retention selections are value-neutral boundary ties without buyer-seller overlap. Horizontal values compare modelled value with hypothetical fixed 50/50, not observed club decisions.

Across 1,500 sensitivity configurations, player-specific selection remains above fixed 50/50, although the difference changes (P10 €5.5M; median €15.7M; P90 €33.6M). In practice, the framework gives clubs an auditable way to compare sell-now and retained-right structures before negotiations, while exposing infeasible regions. It is decision support rather than a reconstruction of actual transfer outcomes. The simulation does not estimate realised proceeds, observed buyer acceptance, or historically optimal contracts; market value is not transfer fee. Stronger validation requires observed contract terms, buyer behaviour, clause enforcement, and realised payouts.

This research was developed in collaboration with SoccerSolver. SoccerSolver currently works with more than 10 football clubs.

Open-source repository: <https://github.com/RupayanHalder39/MITSloanProblem1-academy-transfer-structure-optimization>